

TORONTO BILLY BISHOP, ON, Canada

WMO: 712650

Lat: 43.6275N

Lon: 79.3966W

Elev: 252

StdP: 14.56

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 3.2	(c) 8.1	(d) -9.3	(e) 3.4	(f) 4.0	(g) -3.4	(h) 4.6	(i) 9.7	(j) 36.1	(k) 24.4	(l) 33.3	(m) 23.4	(n) 12.8	(o) 320	(p) 0.574

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 11.7	(c) 83.0	(d) 70.9	(e) 80.3	(f) 70.0	(g) 77.7	(h) 69.1	(i) 74.1	(j) 79.4	(k) 72.5	(l) 77.5	(m) 71.1	(n) 75.7	(o) 11.7	(p) 270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.4	121.0	77.3	70.8	114.5	75.7	69.4	109.0	74.5	37.8	79.4	36.4	77.6	35.1	75.9	82.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
29.7	26.2	23.1	-2.9	89.7	6.0	4.0	-7.2	92.6	-10.8	95.0	-14.1	97.2	-18.5	100.1
			-4.2	77.2	5.7	2.2	-8.3	78.8	-11.6	80.1	-14.8	81.3	-18.9	82.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	48.2	25.8	27.1	33.7	43.7	54.6	64.6	69.9	70.1	63.7	51.5	40.7	31.8
	DBStd	17.35	10.00	8.85	8.45	6.55	6.34	5.67	5.07	4.54	6.19	6.79	6.96	8.10
	HDD50	3041	752	642	507	204	24	0	0	0	1	64	284	563
	HDD65	6588	1217	1062	970	638	330	76	13	9	95	421	729	1028
	CDD50	2386	0	0	2	16	166	437	615	622	413	110	5	0
	CDD65	459	0	0	0	0	7	63	163	167	57	2	0	0
	CDH74	1676	0	0	0	0	32	218	713	562	148	3	0	0
	CDH80	284	0	0	0	0	3	40	148	77	15	1	0	0
Wind	WSAvg	10.7	13.8	13.1	11.9	11.7	9.5	8.4	7.9	8.0	8.8	10.4	11.5	13.0
Precipitation	PrecAvg	33.3	2.4	2.2	2.6	2.7	2.8	2.7	2.9	3.2	3.0	2.5	3.1	3.0
	PrecMax	45.6	4.8	3.9	6.1	4.9	5.9	5.9	5.6	7.7	8.7	5.6	7.5	6.1
	PrecMin	25.5	0.8	0.8	0.5	1.2	1.0	0.7	0.8	0.7	0.4	0.5	0.5	0.9
	PrecStd	4.7	1.1	1.0	1.3	0.9	1.2	1.3	1.2	1.7	1.8	1.3	1.4	1.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.7	47.9	57.3	65.4	78.6	85.2	89.0	85.9	82.1	71.5	58.7	50.4
		MCWB	46.1	42.6	50.3	53.0	64.3	69.4	72.5	72.0	72.2	64.6	53.7	46.9
	2%	DB	44.0	42.9	51.0	59.3	71.9	79.5	83.8	82.0	77.8	67.4	54.6	46.8
		MCWB	42.1	39.6	45.3	49.7	62.5	68.0	70.8	71.5	69.9	61.6	50.9	44.4
	5%	DB	41.0	40.2	47.3	55.6	67.7	75.8	80.6	79.2	75.1	64.3	52.1	44.2
		MCWB	39.0	37.4	42.6	47.9	59.6	66.7	70.3	70.4	68.5	59.6	48.9	41.8
	10%	DB	38.6	37.9	44.0	52.5	64.0	73.0	77.7	77.0	72.8	61.7	50.0	42.0
		MCWB	36.7	35.4	40.2	45.8	56.9	65.6	69.0	69.5	67.2	57.5	46.9	39.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.6	44.2	51.7	56.0	67.7	73.8	76.3	76.1	73.9	67.0	55.4	48.1
		MCDB	47.5	47.3	55.3	61.4	74.3	78.5	83.6	80.9	79.3	70.0	57.1	49.6
	2%	WB	42.5	40.0	46.8	52.1	64.0	71.0	74.1	74.1	71.8	62.9	51.6	44.8
		MCDB	43.3	42.1	49.7	56.8	70.1	76.2	80.0	78.9	76.1	66.2	53.5	46.1
	5%	WB	39.4	37.8	43.3	49.2	60.5	68.9	72.3	72.6	70.0	60.6	49.4	42.2
		MCDB	40.7	39.9	46.2	53.9	66.1	73.8	77.6	77.2	73.8	63.5	51.6	43.9
	10%	WB	36.8	35.7	40.4	46.6	57.9	66.8	70.7	71.1	68.1	58.0	47.3	39.7
		MCDB	38.3	37.5	43.7	51.4	62.8	71.7	75.6	75.4	72.0	60.9	49.6	41.6
Mean Daily Temperature Range	5% DB	MDBR	10.8	10.6	10.9	11.9	13.3	13.1	13.5	11.7	11.8	10.4	9.2	9.0
		MCDBR	10.6	11.6	14.7	17.6	19.6	17.3	17.2	14.2	13.7	13.4	11.8	10.0
		MCWBR	10.4	10.2	11.0	11.3	11.8	9.6	9.2	8.4	8.9	10.1	10.3	9.1
	5% WB	MCDBR	10.7	11.7	14.0	16.1	18.0	15.2	14.7	12.1	12.3	11.9	11.1	9.9
		MCWBR	10.9	10.9	11.4	12.1	12.0	9.4	8.6	8.1	8.7	9.8	10.3	9.8
Clear-Sky Solar Irradiance	taub		0.303	0.324	0.337	0.384	0.414	0.443	0.437	0.423	0.390	0.368	0.322	0.304
	taud		2.404	2.339	2.395	2.296	2.266	2.215	2.258	2.286	2.356	2.389	2.482	2.442
	Ebn at Noon		262	274	286	279	273	264	264	264	265	255	253	250
	Edh at Noon		24	30	33	39	42	44	42	40	34	29	23	22
All-Sky Solar Radiation	RadAvg		447	681	1089	1405	1707	1853	1860	1636	1311	787	493	359
	RadStd		42	61	109	149	182	135	137	83	120	71	62	38

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (52 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page